Tutorial Replication Guidelines

If you are reading this file, you are ready to reproduce the steps described in the paper **"Scenario-Based Model Validation in Asmeta"**.

This document is a restricted version of the `README.pdf` available in the Zenodo replication package. Instructions on how to use the Docker image and how to install Asmeta manually on Windows, Linux, and macOS are omitted here, since the Docker environment is already configured.

For the complete documentation, please refer to the Zenodo replication package.

Scenario-based Validation Tutorial

The basics of scenario-based validation on Asmeta can be replicated as follows in the Eclipse environment:

1. Open the `S1_BasicStart.avalla` in the Eclipse editor.
2. Run the scenario-based validation by pressing the button.

The output of the scenario execution is displayed in the console, and all checks are passed: `Validation terminated without errors.`

To test a failure, run validation for `S2_Failure.avalla`. The output of the scenario execution is displayed in the console:

```
CHECK FAILED: st(JOB1) = RUN or st(JOB2) = RUN or st(JOB3) = RUN at step 1
[...]
WARNING: validation incomplete - some checks failed or errors occurred
```

To test an invariant violation, run validation for `S3_InvariantViolation.avalla`. The console will report an invariant violation in addition to the `WARNING` message: `invariant violation found.`

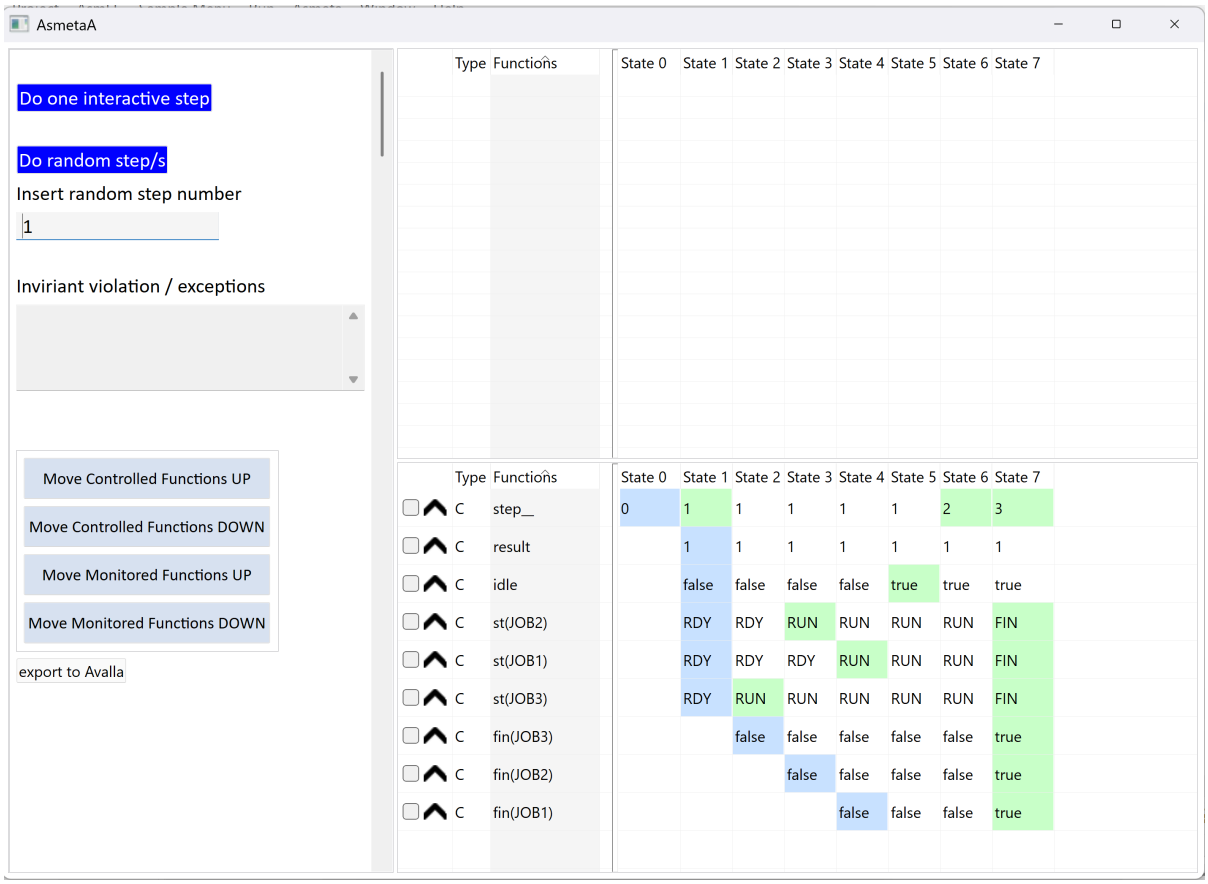
You can also run validation for the scenarios with prefix from `S4_` to `S7_`.

`S8_RandomScenario.avalla` was obtained via random generation (see below). Therefore, since `Scheduler.asm` contains a `choose` rule, the validation can either complete successfully or result in an Action Error:

`SchedulerBlocks.avalla` can be validated and the output will indicate success. However, since it only defines blocks, it is not concretely executed.

Scenario Animation

1. Open the `S5_UsingBlocks.avalla` in the Eclipse editor.
2. Run the scenario animation by pressing the button. The output of the scenario execution is displayed in the animator window.
3. Press `Do one interactive step` button to run the scenario step by step. Press `Do random step/s` button and set the number of steps in the text box under the inscription `Insert random step number` to run automatically more than one step.



Eventual invariant violation (e.g., from `S3_InvariantViolation.avalla`) and errors (e.g., from `S8_RandomScenario.avalla`) will appear in the `Invariant violation / exceptions` box.

You can run scenario animation for all the scenarios with prefix from `S1_` to `S8_`.

Validation with Coverage

1. Open the `S1_BasicStart.avalla` in the Eclipse editor.
2. Run the scenario-based validation with coverage computation by pressing the button.

The output of the scenario execution is displayed in the console. Before the standard success message that indicate that all checks are passed: `Validation terminated without errors`, a structured presentation of coverage data is printed:

```
** Coverage Info: **
** covered rules: **
Scheduler::r_SetFinished()
-> branch coverage:        25.00%
-> rule coverage:          33.33%
-> update rule coverage:    0.00%
-> forall rule coverage:   33.33%
Scheduler::r_Main()
-> branch coverage:       - (no branches to be covered)
-> rule coverage:         100.00%
-> update rule coverage:  - (no update rules to be covered)
-> forall rule coverage:  - (no forall rules to be covered)
Scheduler::r_SetRunning()
-> branch coverage:        50.00%
-> rule coverage:          66.67%
-> update rule coverage:   50.00%
-> forall rule coverage:  - (no forall rules to be covered)
** NOT covered rules: **
```

You can run valiadtion with coverage for all the scenarios with prefix from `S1_` to `S8_`.

Scenario Generation

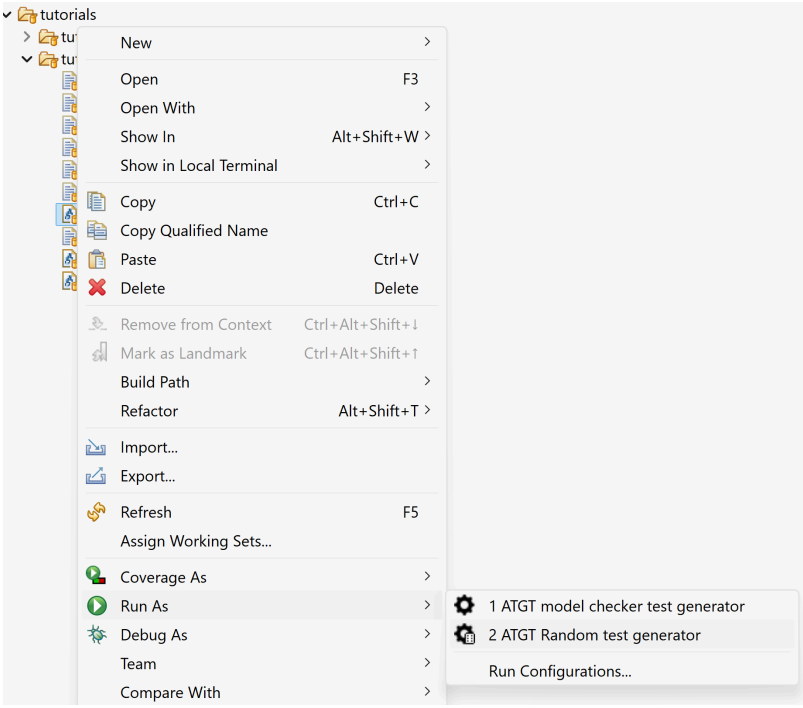
From Animator

1. Open the `Scheduler.asm` in the Eclipse editor.
2. Run model animation by pressing the button. The output of the animation execution is displayed in the animator window.
3. Press `Do one interactive step` button to run an interactive by step, or press `Do random step/s` button and set the number of steps in the text box under the inscription `Insert random step number` to run more than one step with random input.
4. After running some steps, press `export to Avalla` button save the execution as an Avalla Scenario.

The resulting scenario is diplayed on the console and it can be copy/pasted in a new `.avalla` scenario.

Using ATGT random exploration

1. Right click on the `Scheduler.asm` in the Package Explorer window of the Eclipse editor.
2. Select `Run as...` -> `2 ATGT Random test generator`.



The resulting set of scenarios is generated in the `abstracttests` folder.